NELL TRUONG

Software engineering student and game programmer

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ABOUT ME :

I am a software engineering student passionate about building complex systems with a real impact on the users' daily lives. I am seeking a 5-6 month internship or a full-time position in software engineering. My post-graduation work permit application has been submitted.

EDUCATION :

2025 - 2026 | Université de Sherbrooke

Master's in computer science - Game programming (DDJV)

- Courses in-game development, including object-oriented programming, data structures, game AI, computer graphics, distributed systems, multithreading and mathematics.

2023 - 2027 | Toulouse INP - ENSEEIHT

Engineering degree - Software engineering (Equivalent to an MSc in software Engineering)

- Classes in programming, software architecture, software quality, web development, distributed systems, concurrency and parallelism, real-time/embedded systems and formal methods.

2021 - 2023 | Lycée Louis Couffignal

Intensive preparatory class to engineering school

- Intensive courses in mathematics, physics, engineering and computer science.

WORK EXPERIENCE :

Tools programming intern

Ludia | May - August 2026

- Maintained internal tools and added features with a user-centric approach.
- I helped to improve the game engine's editor (C#, WPF, MVVM), particularly on edition tools, navigation and asset management, where I was able to reduce the loading time of sprites from minutes to seconds.
- Fullstack development on the internal LiveOps application (Angular, Java). Added features, unit tests and extended/modernized an existing REST API.

Software engineering intern

Oxydian (Cybersecurity startup) | June - August 2025

- Created APIs, tools and proprietary software solutions for a cybersecurity auditing device.
- Worked with Python, including the FastAPI/Pydantic/SQLAlchemy stack and Docker.
- Implemented automated tests in a CI/CD pipeline on Gitlab.

Game programming intern

SNCF (French national railway company) | June - July 2024

- Worked on a virtual/mixed reality prototype (mainly on hand tracking and the UI) for the Microsoft HoloLens 2 on Unreal Engine 5 (Blueprints & C++), working with a team on Perforce.

PROJECTS :

Mon Ami - 2026

- Online multiplayer horror game developed using Unreal Engine 5.
- Created a Data-Driven procedural dungeon generator.
- Created several gameplay systems including a modular interaction system, a skill and inventory system. All of them designed for multiplayer with server-side validation.
- Developed a discovery system and matchmaking system with an in-game interface and a dedicated C# API deployed on AWS.

Need4Fish / FishEngine - 2025

- Need4Fish is a 3D race against the clock game where the player devours fishes to climb to the top of the food chain.
- During this project, I worked on real-time rendering, culling, 3D game engine architecture, entity component systems (ECS), physics engine integration (JoltPhysics), debugging and CPU profiling in Visual Studio and graphics debugging in RenderDoc.
- Worked on gameplay features like the controls, the first-person camera, the eating/growth mechanism, the user interface and fish AI.
- Developed in a team of four using our own game engine built with DirectX 11 and Jolt Physics.

AI Bootcamp - 2025

- A series of challenges where agents search for exits in a hexagonal grid, solved in a team of four in C++.
- Worked on data structures, problem solving, pathfinding, finite state machines, agent collaboration and interaction, architectural choices for decision-making, Utiliy AI theory and optimization.

N7WS - 2025

- Developed a web application inspired by AWS, for real-time monitoring of our school's computers and to remote script execution.
- Build with Java and Spring Boot (REST API) for the backend and client, a React/TypeScript, Tailwind CSS, RemixJS for the frontend and PostgreSQL for the database, in a team of four.

Tiny scientific calculation model - 2024

- Developed a tool to create visual calculation scripts inspired by Unreal Engine's Blueprints and to transform them into functional Python scripts.
- Developed with a Model-Based Design approach.
- Developed using Java, Gradle, Eclipse Modelling Framework, Sirius and Python.

Want to learn more ? → Check out my Portfolio !

STUDENT SOCIETIES :

TOLOSAT (Student cubesat project)
Flight software team | February 2025 - August 2025

- Worked on a software timer API in C with FreeRTOS.

TVn7 (Audiovisual association)
IT Manager | May 2024 - May 2025

- Tech support, systems administration and maintenance of services.

net7 (Computer science association)
Treasurer | April 2024 - April 2025

- Financial management, fundraising, social media communication and Git workshop instructor.

SKILLS :

Programming

- Git
- C / C++
- Python
- Java
- C# / WPF
- Unreal Engine
- TypeScript
- Svelte/React/Angular

Tools

- Linux
- Docker
- AWS
- Postman

Languages

- French (native)
- English (Fluent)
- Japanese (Beginner)

Soft skills

- Curious
- Team spirit
- Autonomous
- Solution focused
- Willing to learn

HOBBIES :

- Video games
- Board games
- Mountain biking
- Bouldering
- Photography

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